

智能纪要：参考数据模块开发及演示事项讨论

2026年7月8日

🕒 7月8日(周三) 09:42 - 11:20 (GMT+08)

📍 MAN HAIR男士理发店(碧乐城店)

纪要 原始记录

总结

Reference Data Module Development and Demo Preparation

- 💡 **Key Conclusion:**
- Reference data modules are ready for PO demo with consistent UI tabs and maker-checker workflow
 - Load Singapore production data as priority and contact Persona for missing data
 - Migrate only master tables and create new audit logs, keeping implementation simple

Development Progress and Key Decisions

Module Development Status

Modules ready for demo, data sync pending

- Booking location** module completed
- Maker-checker** workflow implemented
- Audit log** functionality added
- Data synchronization from **production** pending
- Test coverage at **88%** for backend

UI Consistency Requirements

Standardize UI tabs and workflow

- Standardize UI tabs: **Draft, Pending, Rejected, Audit**
- First tab shows **approved records**
- Implement consistent **maker-checker** workflow
- Ensure **user profile** integration
- Prepare for **PO demo** next week

Data and Environment Setup

Load production data for demo

- Priority: Load **Singapore** production data
- Fallback: Use **global reference** data
- Contact **Persona** for missing data
- MacBook VPN** issues affecting development
- Deploy to **test environment** next week

Technical Implementation and Audit Log

Audit Log Implementation

Create UI screen for audit log

- Display **before/after values** for changes
- Show **modified fields** clearly
- No need to migrate **old audit logs**
- Keep implementation **simple**
- Use **JSONB column** for audit data

Table Structure and Migration

Migrate only master table (TxRM)

- Migrate only **TxRM master table**
- Do not migrate **TxRH or TxRL** history
- Create new history in **reference data**
- Support **country-specific schemas**
- Use **version 1 parser** for compatibility

Development Priorities

Focus on UI consistency first

- High priority: **UI tab standardization**
- Medium priority: **User profile workflow**
- Low priority: **Production data loading**
- Address **MacBook VPN** issues
- Complete **one module fully** for demo

The audio focused on multiple aspects of the project, including data processing, module implementation, test coverage, user profile, data migration, UI design, and demo preparation. Key decisions were made to ensure consistency, efficiency, and functionality across the system, with a target of presenting a demo to the PO next week.

- **Data Processing and Indexing**

- **JSONB Search Indexing:** When searching within the contents of the JSONB, create indexes to optimize search operations. This ensures that the system can efficiently retrieve relevant data, especially when dealing with large datasets.
- **Data Migration and Synchronization**
 - **Priority of Data Sources:** When synchronizing reference data, prioritize Singapore data. If the required data is not found in the Singapore version, search in the global version. If still not found, consult Persona and CC Johnc.
 - **Master Table Migration:** Only migrate the TxRM master table from production data into the reference data. Do not migrate TXRL and TXRH from the old system, but ensure that the workflow related to these tables is properly implemented and visible to the customer.
 - **Data Synchronization Script:** A data migration script has been created to synchronize the table from imported data into the application against the database. Consider using a reconciliation script for handling new updates or recreating the table when necessary.

- **Module Implementation and Testing**

- **Module Implementation Time:** The implementation of a specific module took 3 days. This was due to the need to adjust functionality based on the production data structure, conduct research on parameters, and make changes to the previous implementation to ensure compatibility.
- **Test Coverage**
 - **Backend Coverage:** The backend test coverage for the module reached 88%. This indicates a relatively high level of testing for the backend functionality, ensuring the stability and reliability of the system.
 - **Front - End Coverage:** There is a need to increase front - end test coverage. Currently, the front - end only handles AOV tests, and efforts should be made to improve the overall test coverage for the front - end.
 - **Unit Testing Standardization:** Standardize the unit testing framework across all modules. Use only one framework for consistency, whether it is for the front - end or the back - end. For the back - end API testing, use the G unit framework.

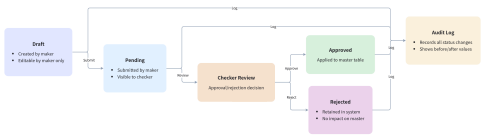
- **User Profile and Workflow**

○ **User Profile Integration**

- **Local and QA Environments:** The user profile has two implementations. In the local development environment, it is set up in the local database, while in the QA environment, it is integrated with the CSM Module. Ensure that the user profile setup is consistent across different environments and modules.
- **Maker and Checker Workflow:** Implement the maker and checker workflow in the user profile module. The maker creates a draft, and the checker approves or rejects it. This workflow should be consistent across all reference data modules.

○ **Workflow Standardization**

- **Consistent Status Display:** Standardize the display of workflow statuses such as draft, pending, reject, and audit log across all modules. These statuses should be visible to the user in a consistent manner, regardless of the module they are using.



Maker-Checker Workflow

- **Query Lookup:** The query lookup function is not necessary for all modules. It is an optional feature that may be specific to certain modules and is not required for overall system consistency.

● **UI Design and Consistency**

○ **Tab and Status Standardization**

- **Five - Tab Structure:** Standardize the UI to have five tabs: record (or the module - specific name), draft, pending, rejected, and audit (previously called history or audit log). This provides a unified view for the user across different modules.
- **Status Number Display:** Display status numbers for draft, pending, and rejected items. Do not display numbers for audit and approved items, as they are not part of the approval process. This helps the user quickly understand the status of the data at a glance.

Tab Name	Description	Status Display	Number Display
Record (Module-specific name)	Shows approved records Main operational view	Approved items only	No
Draft	Maker's unsubmitted edits Private to creator	Draft items	Yes
Pending	Awaiting checker approval	Pending approval items	Yes
Rejected	Checker-rejected submissions	Rejected items	Yes
Audit	History of all status changes	Before/after values Change timestamps	No

Tab must be consistently named and ordered across all reference data modules to ensure user familiarity.

Standardized UI Tab Structure

- **Search Functionality:** The search function is not required for all modules. It can be included based on the data volume and user needs of each module. For example, if a module has a large amount of data, a search function can be added to improve user experience.

- **UI Style:** Do not force standardization of the UI style, as different users may have different preferences. However, ensure that the overall workflow and status display are consistent.
- **Demo Preparation**
 - **Showcase for PO:** Prepare a demo for the PO next week. The demo should include the user profile, workflow, and audit log features. It should demonstrate the consistency and functionality of the system across different modules.
 - **Highlight Workflow:** Emphasize the maker and checker workflow, as well as the consistent display of draft, pending, reject, and audit log statuses. This will give the PO a clear understanding of the system's operation.
- **Technical Issues and Solutions**
 - **MacBook Policy Issues:** The SCB macOS policy on MacBooks causes issues with installing extensions in VS Code, accessing Github when connected to the VPN, and using Copilot. Track this issue with the MacBook support team and provide an Excel list of the problems to facilitate resolution.
 - **Audit Log Structure:** Create a parser for the v1 audit log structure, which is different from the current system's structure. However, this is a lower - priority task as there is currently no requirement to display the old audit log.
- **Data Source and Requirements**
 - **User Profile Data:** The user profile data in the global security schema is currently empty. Request the data from Persona and CC Michael and John if it is not provided in the zip file from Prasana.
 - **Business Requirements:** The primary objective of the reference data system is to support bank guarantees. Ensure that the system can load production data, especially the master table, to support this functionality.
- **Subsequent Work Arrangements**
 - **Data Import and Module Completion**
 - Master Data Import: Import master data into the system before next week. However, due to the workload, prioritize tasks based on importance. If necessary, postpone database setup for some modules.
 - Module Completion: Complete the remaining modules and ensure that they are ready for the demo next week. Coordinate with team members to ensure smooth progress.
 - **Email Summary and Synchronization:** Compile the meeting items into an email and send it out. Schedule a short meeting in the afternoon to go through the email and synchronize the progress.

- **Demo Participation:** Although some team members may be on leave, ensure that there are presenters available for the demo next week. Coordinate with the presenters to ensure a successful demonstration.

👉 智能纪要反馈收集 [该内容不支持导出查看]

智能章节

00:00 Discussion on JSONB index configuration and the time cost of original implementation conversion

This meeting section covers two key points: first, the note that indexes should be created for JSONB content if inner-content search is required, and second, the conversion of the original implementation took 3 days. The 3-day timeline included adjustments to fix previous incompatible structure issues, supplementary research on production data, historical business logic, and relevant books to clarify unstated BRD rules and actual business demands.

02:01 Discussion on Test Coverage Status, Testing Framework Consistency and Subsequent Demo Arrangement

This meeting section centers on test coverage updates: the backend country part has reached 88% test coverage, frontline only conducts AOV tests while the overall test coverage is around 80%. Participants confirm they will unify the testing frameworks across modules, using the consistent specified tool for frontend and standard JUnit for backend API tests. The session then prepares to move on to Michael's demo next.

04:02 Alignment on Consistent Maker-Checker Workflow and Relevant Module Implementation Arrangements

This meeting section confirms that the default draft-approval flow of the flow director should be implemented consistently across all reference data modules. James is required to model this flow on the event modeling board for shared understanding by users and POS. John C, Jimmy shall collaborate with Neil and Clement to build the maker-checker mechanism into the user profile module, which currently has local dev environment and QA CSM integrated two implementations. The team will align related rules and prepare real recognizable reference data for upcoming product demo.

08:55 Sharing Practice of Fuzzy Table Search in Multi-schema Massive Database

This meeting section focuses on Michael's data lookup experience. He encountered heavy difficulty in finding required master-related tables from the database, as the SGS schema alone has nearly 400 tables with another 200 additional tables. He later used a convenient new skill that performs fuzzy search across all database schemas, conducts tokenization,

matches table and column names, scores results, and exports qualified data such as booking location information to Excel, which avoids previous repeated rework.

12:11 Progress Update on Booking Location Feature Development and Data Synchronization Plan

This section centers on speaker 4's work progress update: they developed the booking location feature, whose table structure aligns with the personas database, and related amending work is nearly finished. They noted pending validations, uncompleted database synchronization due to difficulties locating the charge part, and will implement the synchronization right after the meeting, having reached a consensus with John on text and customer onboarding issues yesterday. The ongoing text test is also close to completion.

14:30 Audit Log Feature, UI Functionality and Related Data Logic Discussion & Demo

This section mainly discusses the audit log related features: it records data changes with before and after values, stored as a plain JSON column without indexes for now, supports JSON output. The relevant UI adds pagination, sorting and search functions, as the data is imported from persona database, and also involves related data validation for approval flow, SCD and currency related change scenarios.

18:30 Discussion on Data Migration, Synchronization Scripts and Related Module Progress

This session centers on data migration related work. The team has developed a data migration script to sync tables between imported data and the application database. They discussed the necessity of an extra reconciliation script, noting it is unnecessary when the two tables are identical, while it can be adopted for full table recreation after new updates are obtained. The migration script supports automatic running and shell invocation. Work on booking location module is paused, and data synchronization for bank detail, bank holiday modules is still ongoing.

20:23 Demo of the Module with Draft and Audit Log Features

This section is a demo by Speaker 6 for a module with draft and audit log features, showcasing its full workflow. The maker can save drafts repeatedly before submission, while drafts are only visible to the maker and hidden from checker-level supervisors, who can view pending submissions to approve or reject. The 4 core operations include creating, updating, submitting the draft and getting it approved, with audit logs recording changed fields, action status, timestamps and related info, and Speaker 6 is about to introduce the second flow.

24:22 Cross-module Function Standardization, Technical Solution Alignment and Next Week Demo & Deployment Preparation Discussion

This meeting section focuses on standardizing cross-module workflows, requiring all modules to apply draft, pending, reject statuses, audit logs and maker-checker rules, plus unified JSON data formats and shared processing utilities. The team notes version compatibility risks, plans

test data preparation for next week's demo, follows up on dedicated dev environment access, and confirms a shrink-based branching strategy for all owned modules.

35:35 Introduction and Demo of PR Check Coverage Script, Follow-up Arrangement of Remaining Meeting Time

This chapter mainly sees Speaker 6 briefly introducing a draft PR check script for backend tracker code coverage verification, which has Python and upcoming shell versions, can be triggered automatically or run manually, checks if coverage meets the 80% threshold, blocks unqualified pushes, and is to be further reviewed and discussed with Josie. The attendees then note the limited remaining meeting time, as Speaker 1 needs to leave for another meeting soon.

39:43 Discussion on Table Schema Version Synchronization Logic for Singapore and Global Editions

This chapter mainly covers the development approach for query preparation related to table schemas. Two versions of table schemas are obtained from the Prasana side, respectively the Singapore version and the global version. For reference data synchronization, the team prioritizes matching with the Singapore version first, and will search the global version if no corresponding result is found in the Singapore version, while another participant comments that this approach is random.

40:26 Discussion on Audit Log UI, Master Table Migration and Parallel System Operation Arrangement

This meeting focuses on two core tasks: developing a consistent audit log UI that displays modified fields and full edit history, and handling master table migration for txrm. The team decides to exclude old system history/non-txrm data for now, adopt the simplest implementation without unconfirmed BO history requirements. The new reference data system will run in parallel with the existing production system, serving the primary bank guarantee scenario first.

51:35 Discussion on Issues Caused by SCB Mic OS Policy for MacBook and Follow-up Action Arrangement

This section discusses issues caused by the SCB mic OS policy on company MacBooks: users cannot auto-install VS Code extensions, access GitHub or use Competitor in VS Code when connected to VPN, while they cannot access company internal websites offline. Speaker 3 will follow up with MacBook support for fixes, and both speakers agree to collaborate on sorting out an Excel list to clearly categorize function availability under VPN-connected and disconnected scenarios for subsequent problem solving.

53:27 Discussion on Audit Log Parser Requirement and User Profile Data Acquisition

This session covers two core points: First, the team's audit structure only retains changes instead of copying full JSON before and after (saving DB space), but a low-priority v1

production audit log parser is needed for potential future log display. Second, regarding the required FAB5 functional access and Tax Beta access related to production user profiles, the team found the relevant data table is under global security schema but currently empty, and will follow up with Prasana to get required data, with copies to Michael and the speaker if needed.

56:32 Discussion on Upcoming Appeal Show-and-Tell Preparation

This meeting section confirms the appeal show-and-tell will be held in two weeks, and more manual information is to be prepared for it. The materials to present include existing contents, Cody Code, uniform checkers that deliver consistent user experience across different reference data scenarios, the audit log, and all previously presented modules. The team member also notes that others can ping him for questions, and call him directly if he does not reply to urgent matters.

58:04 Meeting discussion on upcoming Knox's demo arrangement and table schema synchronization rules confirmation

This section confirms two key points: first, Knox will deliver a demo of the current completed work to PO members Akite and Malaysian Chinese BG PO Cindy next week. Second, the table schema synchronization priority rule is set as Singapore first, then global version, and if neither works, email Persona with Jongc copied as Johnc suggested.

01:00:14 Discussion on Developing and Displaying Audit Log UI Screen

This meeting section focuses on the audit log related arrangement: it is required to create the corresponding UI screen to display the audit log. The content presented on the UI screen should strictly follow the previously discussed specifications, which refers to the content that Peng demonstrated in the meeting today.

01:00:46 Discussion on Master Table Data Migration Scheme and Reference Data Workflow Confirmation

This section confirms the master table migration rule: only TxRM will migrate and load production data to the reference table, while TXRH and TXRRL will not load production data there. The team confirms they will keep the existing current workflow structure which has an intermediate table to cache pending drafts and relevant audit info, no structure adjustment is needed, and the final confirmed approach will be shared via email later.

01:04:00 Discussion on Unifying UI Tab Naming and Workflow Rules Across Modules for Consistency

This meeting focused on unifying cross-module UI workflow per Knox's "consistent" requirement. 5 uniform tabs are confirmed: module master (approved records), draft, pending, rejected, audit. Unnecessary query lookup and current search functions are abandoned. Draft shows personal drafts only, pending displays all pending entries. Reject re-editing scenario is left for follow-up module confirmation.

01:19:24 Discussion on Unifying the Display Rules of Tab Statistics Numbers Across Modules

This discussion focuses on unifying the inconsistent quantity display numbers on module tabs of the system. Participants agree to exclude non-approval status tabs like audit, determine that only relevant approval status tabs need to show the counts, and note that most existing modules already have such numbers but lack uniform standards, reaching a general consensus to align the display rules later.

01:23:09 Discussion on UI style consistency and search function design for different modules

This section focuses on UI style consistency and search function arrangement of product modules. Participants discuss that previous UI adjusted per PR review comments turned out unsatisfactory, but they decide not to unify diverse existing UI styles temporarily as preferences of PO and business teams are unclear. They also agree that the search function should be set based on actual data volume, which is necessary for modules with large data and unnecessary for modules with small data such as the holiday module.

01:25:40 Discussion on Task Prioritization and Master Data Import Arrangement for Upcoming PO Demo

The team discussed task constraints: some members lack time to finish remaining models and import master data before the deadline, local databases are not fully set up, and the master data import script is not universal. They decided to prioritize tasks since the upcoming PO demo only needs to present surface contents. The lead will sort out all task items, send them via email, and arrange a short follow-up check later.

01:30:02 Discussion on Development Environment Platform Availability and Copilot Connection Issues

The team discusses device platform issues: they haven't confirmed if Windows can be used for their work, worry about redundant deployment work if building on Windows first then switching to required high-spec Mac devices, note that the database has been imported to Windows, and point out that GitHub Copilot is unstable under VPN connection which would interrupt code running. They plan to first submit the device list and clarify the Windows availability issue later.

01:31:28 Discussion on multi-country data schema design and follow-up arrangement for country visitor system

This section discusses the country visitor system's data architecture, confirming the new system will follow the production logic that each country corresponds to an independent, separate database schema, which is not in the current code scope. The login function that lets users select their corresponding country to access exclusive data is put on hold temporarily

and will be discussed after the Singapore iteration is finished. The meeting will follow up via a sorted email and a short afternoon sync session.

01:34:19 Discussion on the Upcoming Demo and Related Arrangements

This meeting section covers that Speaker 3 will be absent for the next week's demo to PO, with Yue Ge and Gang Ge taking charge, and Speaker 3 will treat everyone to a meal after his vacation. The team notes PO is not urgent for the demo, and the actual demo meeting with PO is not confirmed yet. Knox hopes the demo can show the team's work to inspire PO and boost their confidence, to help push PO forward.

👉 智能纪要反馈收集 [该内容不支持导出查看]

关键决策

- **Key Decision:** Standardize the UI tab structure for all reference data modules to be consistent for the upcoming PO demo next week.
 - **Problem:** The UI workflow tabs across different reference data modules are inconsistent, which will cause confusion for users and PO during the demo.
 - **Discussion Plan:**
 - Speaker 3 proposed that all modules should unify the UI tab structure to ensure consistent user experience.
 - Speaker 2 suggested that the core workflow tabs (draft, pending, rejected) should be fully consistent, while the first tab name can be adjusted according to the module's actual business.
 - The team agreed to name the history/audit related tab uniformly as "audit" to avoid overly technical naming.
 - **Decision Basis:** Consistent UI structure can reduce user learning cost, and meet the requirement of unified workflow display for the PO demo.
- **Other Decisions:**
 - a. When synchronizing table schema, prioritize Singapore version first, then global version if not found.
 - b. Only migrate TxRM master table from production data, no migration for TXRH and TXRL tables.
 - c. The team will prepare for the PO demo scheduled for next week.
 - d. Speaker 3 will sort out the confirmed items and send a synchronization email to the team.
 - e. The query lookup function is an optional module, not required for all modules.

金句时刻

「So let's keep it simple.」

—— This core principle guides the team to avoid over-planning unrequired functions, which improves development efficiency.

「as a user, they definitely hope that the UI template is unified across all modules」

—— This statement clearly states the core value of UI standardization from the end user perspective.

「Knox hopes to give PO some inspiration or confidence through the demo, so that he can push the PO side to move forward」

—— This clarifies the actual purpose of the upcoming PO demo, helping the team align on the demo focus.